

Implementation Plan: Interactive 3D Haunted House Simulation

Project: Simple OpenGL Project | Roll 2107091 | Student: MD.SABITH **Stack:** C++, OpenGL 1.x (fixed-function), GLUT/FreeGLUT, GLU quadrics **Deliverable:** one source file (`haunted_house.cpp`), no external models or textures

1. Goal and Scope

Build a walkable, first-person horror estate inspired by *Granny* and *Alan Wake 2*:

- A fenced compound (80 x 80 units) with a working gate, cobblestone path, dead trees, graveyard, wagon and lamp posts.
- A stone mansion with a round tower, timber turret, porch, entrance hall, staircase, gallery and three bedrooms.
- A curved railway with an animated train in the forest outside the east fence.
- Eight fixed-function lights, zone-based fog, lightning, and continuous animation.

Constraints: primitives only (cube, prism, cylinder, sphere, cone, torus, quads), OpenGL 1.x, must run smoothly on ordinary lab PCs.

2. Coordinate System and Layout

X points right, Z points toward the viewer, Y is up. 1 unit is about 1 metre.

Zone	Position	Contents
Fenced compound	X, Z in [-40, 40]	Fence, pillars, gargoyles, gate at Z = +40
Path	X in [-1.5, 1.5], Z = 60 to -11.5	Cobblestone quads, five porch steps
Mansion	X in [-12, 12], Z in [-30, -14]	Hall, gallery, three bedrooms, roof
Tower / turret	(-13.5, -11.8) / (9, -27)	Round stone tower, leaning timber turret
Graveyard	X in [-38, -22], Z in [12, 34]	Tombstones, crosses, ghost
Railway	$X = 55 + 8 \sin(Z/25)$	Rails, sleepers, train
Forest	Outside the fence	Pine trees, thicker fog

Key levels: ground floor at Y = 0.3, upstairs floor at Y = 4.5, wall height 9, eye height 1.7.

3. Architecture

```
main()      -> glutInit, window, callbacks, init(), glutMainLoop
init()      -> GL state, fog, quadric, star field, generate tree spots,
              door setup, build all display lists
display()   -> flashlight (eye space) -> gluLookAt -> world lights -> sky
              -> static lists -> dynamic objects -> transparent objects -> HUD
idle()      -> dt from elapsed time -> update() -> redisplay
update(dt)  -> look, doors/gate, walking + collision, floor height,
              fog zone, flicker, lightning
input       -> keyboard up/down, arrows, passive mouse motion
```

Design rules

1. Static geometry goes into **display lists** (ground, path, fence, trees, pines, graveyard, props, house exterior, house interior, rails, gate leaf).
 2. Anything that moves or flickers is drawn **each frame** (doors, gate, chandelier, flames, drapes, rocking chair, train, ghost, bats, crows).
 3. All motion uses **delta time**, so speed does not depend on the computer.
 4. Transparent objects are drawn **last** with depth writing off.
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4. Build Order (Milestones)

Each step produces something you can run and check before moving on.

Step	Work	Check
1	Window, projection, first-person camera, ground, sky, moon	You can look around a dark field
2	Primitive helpers: box, cylinder, cone, sphere, prism, subdivided box	Test shapes render with correct lighting
3	Fence posts and rails, iron pickets, pillars, gargoyles, gate, path	Complete fence around the estate
4	Mansion walls, roof, gables, dormers, chimneys, tower, turret, porch	House looks right from the gate
5	Dead trees, graveyard, wagon, pumpkins, lamp posts	Yard looks abandoned
6	Hall floor, staircase, balustrade, gallery, chandelier, portraits	Stairs are visually correct
7	Three bedrooms and furniture	Each room matches its reference
8	Railway, sleepers, locomotive, carriages, pine forest	Train loops along the curved track
9	All eight lights, materials, per-zone light enabling	Each light can be identified on its own
10	Doors, gate, ghost, bats, crows, drapes, flicker animations	Everything animates continuously
11	Collision, stair height, fog by zone, lightning, HUD	Cannot leave the estate or pass through walls
12	Testing, tuning, performance pass	Smooth on lab hardware

5. Module Details

5.1 Primitive helpers

- `box`, `cyl`, `cone`, `sph`, `ell`: wrappers using `glutSolidCube` with scaling, `gluCylinder`, `glutSolidCone` and `glutSolidSphere`. Cylinders and cones are rotated so they point up (+Y).
- `prism` / `prismZ`: custom triangle and quad prism for roofs, with a ridge along X or Z.
- `bigBox`: a box whose faces are split into cells (about 1 unit). Each cell gets a small random brightness. This gives a stone or plank look, and it lets the fixed-function lights work per cell instead of per huge face (important for the flashlight).

- `wallBox`: draws a wall and registers its collision box in one call.

5.2 Boundary and grounds

- Wooden sections use posts every 4 units (about 80 around the compound) with upper rail at 1.6 and lower at 0.7. Some are tilted.
- The front uses iron pickets (spacing 0.25) with rails, stored in a display list.
- Stone pillars at corners and beside the gate. Gargoyles on the gate pillars.
- Two gate leaves hinge at $X = \pm 4$ and swing up to 70 degrees.
- Ground is a subdivided quad grid (finer inside the fence) with grass tufts made of thin cones. The path is a grid of quads with random brightness.

5.3 Mansion exterior

- Outer walls are thin boxes (0.4 thick) with a door gap in the front wall, plus a darker lower band.
- Roof prism with ridge at height 12, two front cross gables, four dormers, three chimneys of different heights.
- Round tower (cylinder plus cone roof plus spire, arched slits), tilted timber turret (stacked boxes, leaning top block).
- Porch with two columns, lintel, prism roof and five step cubes.
- Windows are emissive quads drawn without lighting: about one third lit amber, the rest dark and cracked.

5.4 Interior

- **Hall**: plank floor with missing and raised planks, green-grey lined walls with peeling patches, ceiling beams.
- **Staircase**: 14 stacked cubes (rise 0.3, run 0.45, width 4), balustrade posts every step, sloped handrail, newel posts.
- **Gallery**: upstairs slab with a front rail; three bedroom doors open onto it.
- **Decor**: chandelier (torus ring, six arms, flames), candelabras, six portraits (one crooked).
- **Bedroom A (R9)**: plank walls, bed, rocking chair, dresser, radiator, hanging lamp, barred windows.
- **Bedroom B (R10)**: dark bed, torn drapes, bedside candle, rags, very low light.
- **Bedroom C (R11)**: two iron beds, canopy drape, wardrobe, striped wallpaper, holes in the floor, dust.

5.5 Railway and forest

- Track follows $x = 55 + 8 \sin(Z/25)$. Rails are built in 2-unit segments rotated to the curve tangent; sleepers every 0.7 units on a gravel strip.

- Train head position loops over 30 seconds. Each car is placed on the curve and rotated to match its own heading.
- Locomotive: chassis, body, nose, cab, roof, chimney, rust patches, emissive headlight. Wheels spin with train speed.
- Pines are stacked cones on cylinder trunks, placed in rows on both sides of the track plus a ring around the fence.

5.6 Lighting (8 lights)

Light	Type	Purpose
LIGHT0	Directional (w = 0), cold blue	Moon; dimmed smoothly when indoors
LIGHT1	Spot, cutoff 22, set in eye space	Flashlight on camera (F toggles)
LIGHT2	Point, warm amber, flicker	Chandelier
LIGHT3	Point, orange, strong flicker	Bedroom B candle
LIGHT4	Point, warm yellow, occasional flicker	Lamp post at the gate
LIGHT5	Spot, moves with the train	Train headlight
LIGHT6	Point, low range	Window glow in front of the house
LIGHT7	Point, pale blue, follows the ghost	Ghost glow

Only lights within range of the player are enabled each frame. World lights are positioned after `gluLookAt`. The flashlight is positioned before `gluLookAt`, while the modelview matrix is identity.

5.7 Atmosphere

- `GL_EXP2` fog with teal colour. Density eases toward 0.006 inside, 0.018 outdoors, 0.03 near the railway. The clear colour matches the fog colour.
- Sky drawn with fog and lighting off: gradient walls, 200 stars, moon with halo, slow cloud quads.
- Lightning: random timer, double flash of about 0.12 s that brightens sky, ambient and fog. L triggers it manually.
- Blob quads under trees stand in for shadows.

5.8 Animation table

Object	Motion
Front door, bedroom doors	Rotate about the hinge; E toggles; first bedroom door creaks open when near
Gate	Opens when the player is within 7 units; E flips the state
Ghost	Drifts on a path, bobs, alpha $(0.35 + 0.15 \sin(0.7t))$
Bats / crows	Orbit paths; wing angle from a sine wave
Chandelier	Pendulum swing of $3 \sin(1.1t)$ degrees
Candles, lamps	Flicker from sine mixes plus small random values
Drapes	Vertex offsets from a sine wave
Rocking chair	Rocks about its base
Train	Position along the curve, wheels turn, loops every 30 s

5.9 Camera, controls and collision

- Camera at eye height 1.7 with yaw and limited pitch. Mouse look uses `glutWarpPointer`.
- Movement: 3 units/s, 6 with Shift, plus a small head bob.
- **Collision:** the player is a circle (radius 0.35) tested against axis-aligned boxes; each box has a Y range, so ground-floor and upstairs walls do not interfere. The player slides along boxes. Closed doors and the closed gate are temporary boxes each frame.
- **Floor height:** a function returns porch step height, stair height under the player, gallery/bedroom floor upstairs, or ground. Camera height eases toward it.
- Viewpoint keys 1 to 4 teleport to gate, hall, bedroom, railway.

6. Performance Plan

- Static parts in display lists (one call each).
- Only nearby lights enabled; fog hides the 400-unit far plane.
- Low slice counts for distant or small shapes (pines use 8 slices).
- Immediate-mode drawing kept for a small number of dynamic objects.
- Target: 60 fps at the start position and inside the hall.

7. Testing and Verification Checklist

- ☐ Program compiles with no errors on the lab compiler.
 - ☐ Runs at about 60 fps at start and inside the hall.
 - ☐ Player cannot leave the fence, pass through walls or fall through stairs.
 - ☐ Gate, front door and bedroom doors open and close with E.
 - ☐ All eight lights can be identified by isolating them; F toggles the flashlight.
 - ☐ Fog density changes between yard, house interior and railway.
 - ☐ Viewpoints 1 to 4 match R3/R2, R6, R9 to R11, and R12/R13.
 - ☐ Train loops every 30 seconds and its headlight lights the track.
 - ☐ Lightning flashes at random and with L.
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8. Risks and Mitigations

Risk	Mitigation
Fixed-function lighting is per vertex, so spot and point lights look flat on large faces	Subdivide big surfaces into cells (<code>bigBox</code> , grid floors)
More than eight lights needed	Reuse the eight slots and enable only nearby ones
Z-fighting between wall linings, windows and planks	Offset decorations 0.02 to 0.1 units from the surface
Player stuck in corners	Circle-vs-box push-out with three iterations per frame
Slow on weak GPUs	Display lists, culled lights, low-poly cones for forest
No real shadows in OpenGL 1.x	Dark blob quads under trees and props

9. Future Work

- Sound (footsteps, door creak, train) with a small audio library.
 - A game goal in the style of *Granny*: find a rusty key in Bedroom B to open the gate.
 - Procedural or image textures for stone, planks and cobblestones.
 - Shaders for real shadows and volumetric fog (outside the OpenGL 1.x scope of this project).
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10. How to Build and Run

```
bash
```

```
# Linux
```

```
sudo apt install g++ freeglut3-dev
```

```
g++ -O2 -o haunted_house haunted_house.cpp -lglut -lGLU -lGL -lm
```

```
./haunted_house
```

```
# Windows (MSYS2 MinGW64)
```

```
g++ -O2 -o haunted_house.exe haunted_house.cpp -lfreeglut -lglu32 -lopengl32
```

Controls: W A S D move (Shift run), mouse or arrows look, F flashlight, E door/gate, L lightning, 1 to 4 viewpoints, H help, M mouse look on/off, Esc quit.